Chronic lymphocytic leukemia / small lymphocytic lymphoma

CD5+ CD23+ B-cell neoplasm · **treat only on an iwCLL indication** · **TP53 + IGHV** pick between continuous BTKi and fixed-duration venetoclax · chemoimmunotherapy now rarely used

Compiled from iwCLL 2018 · NCCN · WHO-HEM5/ICC 2022 · primary trial reports verified against PubMed, October 2026.

DIAGNOSIS — CLL VS SLL VS MBL (IWCLL 2018, WHO-5)

Entity	Defining features
CLL	**≥5×10^9/L clonal B cells in blood, sustained ≥3 mo**; clonality by light-chain restriction on flow. A cytopenia from typical marrow infiltrate = CLL at any count
SLL	Same disease, nodal: **nodes and/or spleen, blood B cells <5×10^9/L**, no marrow-driven cytopenia; confirm on **lymph node biopsy**
MBL	Clonal B cells **<5×10^9/L**, no nodes, organomegaly, cytopenia or symptoms. **Low-count <0.5** (rarely progresses) vs **high-count 0.5–4.9**. Progression to treatment-requiring CLL **~1–2%/yr**; more infections and skin cancers
Prolymphocytic progression	WHO-5: **>15% prolymphocytes** in blood/marrow. **WHO-5 dropped B-PLL** (now CLL with PL progression, MCL variant, or splenic B-cell lymphoma with prominent nucleoli); **ICC 2022 keeps B-PLL** narrowly
Richter transformation	Aggressive lymphoma arising in CLL — DLBCL type (~90%) or Hodgkin type. Page 2

FLOW PHENOTYPE AND DIFFERENTIAL

- **CD19+ CD5+ CD23+** with **dim CD20, dim CD79b, dim surface Ig** (single light chain). **CD200+**, LEF1+, ROR1+; FMC7–. Smudge cells on smear
- **Matutes (RMH) score:** 1 point each for CD5+, CD23+, FMC7–, weak sIg, weak CD22 (CD79b in the modified score). **4–5 = CLL** (87% of CLL); 3 = atypical; 0–2 = other B-LPD
- **Mantle cell is the dangerous mimic** (CD5+): **CD23–, CD200–, bright sIg/CD20, cyclin D1+, SOX11+, t(11;14)**. Atypical/CD23-dim cases: **cyclin D1 IHC or t(11;14) FISH**
- Others: marginal zone (CD5–, CD23–), hairy cell (CD11c/CD25/CD103, BRAF V600E), follicular (CD10+)

PRETREATMENT WORKUP (IWCLL)

- **Before every line of therapy: FISH** (del13q, del11q, +12, **del17p**) **+ TP53 sequencing** — clones evolve; TP53 status flips the regimen
- **IGHV mutation status once** (doesn't change): **unmutated = ≥98% identity to germline**; mutated = >2% deviation
- **CpG-stimulated karyotype:** complex = ≥3 abnormalities; **≥5 is independently adverse**; 3–4 adverse mainly with TP53 aberration
- CBC + smear, CMP, LDH, uric acid, **β2-microglobulin**, **DAT** + retic/haptoglobin, **quantitative IgG/IgA/IgM**, **HBV (HBsAg, anti-HBc), HCV**, HIV
- **CT neck/C/A/P before venetoclax** (TLS risk) or in trials; not needed for watch-and-wait. **PET only if Richter suspected.** Marrow not required for diagnosis — do it for unexplained cytopenias
- Cardiac history, ECG, BP, anticoagulants/antiplatelets and CYP3A drugs before choosing a BTKi

TREATMENT PATHWAY

Diagnosis → is there an iwCLL indication?

No → watch and wait
CBC q3–6 mo; treating early does not help:
CLL12 ibrutinib vs placebo — **EFS gain, no OS**

Yes → test TP53 (FISH + seq) and IGHV
then fitness, cardiac risk, drug interactions, and patient preference: fixed vs continuous

First line — TP53 aberrant (del17p / TP53mut)

Continuous covalent BTKi
zanubrutinib or **acalabrutinib** (± obinutuzumab)
SEQUOIA arm C: 18-mo PFS 88.6%
fixed-duration regimens relapse earlier here
no chemoimmunotherapy

First line — no TP53 aberration

Fixed duration — time off therapy
VenO ×12 cycles (CLL14; CLL13 in fit)
acalabrutinib-venetoclax ×14 (AMPLIFY)
ibrutinib-venetoclax ×15 (GLOW; not US-labeled)

or continuous BTKi
zanubrutinib · **acalabrutinib ± O**
pirtobrutinib (1L label Oct 2026, no del17p)
ibrutinib: effective, more cardiac toxicity

Chemoimmunotherapy — niche
FCR only IGHV-mutated, fit, young — rarely now
targeted regimens beat it (FLAIR, CLL13)

Relapse

Progressed on a covalent BTKi
VenR ×24 mo (MURANO) or **pirtobrutinib**
don't switch cBTKi→cBTKi for progression
(switching for intolerance is fine)

Relapse after fixed-duration venetoclax
long remission: **retreat venetoclax-based**
short remission: **covalent BTKi**

Double-exposed (BTKi + BCL2i)
pirtobrutinib · **liso-cel** · trials
allo-HCT if fit — the only curative option

IWCLL INDICATIONS TO TREAT — NEED ≥1

- **Progressive marrow failure:** Hb **<10 g/dL** or platelets **<100×10^9/L** (a stable plt just under 100 may be watched)
- **Spleen ≥6 cm below costal margin**, or progressive/symptomatic · **nodes ≥10 cm**, or progressive/symptomatic
- **Lymphocyte doubling time <6 mo** or ≥50% rise in 2 mo (ALC <30: observe longer before calling LDT)
- **Autoimmune anemia/ITP poorly responsive to steroids** · symptomatic extranodal disease
- **Constitutional:** weight loss ≥10% in 6 mo · fatigue ECOG ≥2 · fever ≥38.0°C ≥2 wk · night sweats ≥1 mo (no infection)
- **NOT indications:** high ALC alone (leukostasis is rare), hypogammaglobulinemia, a paraprotein, adverse FISH/IGHV, Rai stage alone

STAGING

System	Definition · survival
Rai	**0** lymphocytosis only (low) · **I** nodes · **II** spleen/liver (intermediate) · **III** Hb <11 · **IV** plt <100 (high). Original median survival >150 / 101 / 71 / 19 / 19 mo
Binet	Areas: head-neck, axillae, groins, spleen, liver. **A** ≤2 areas (~55%; near-normal survival) · **B** ≥3 areas (~30%) · **C** Hb <10 or plt <100 (~15%; median ~2 yr)

Anemia/thrombocytopenia from **autoimmune destruction does not upstage** — Rai III/IV and Binet C mean marrow failure.

CLL-IPI (2016)

Factor	Pts
TP53 aberrant (del17p and/or TP53 mutation)	**4**
IGHV unmutated · β2M >3.5 mg/L	2 each
Rai I–IV or Binet B–C · age >65	1 each

Low 0–1 · intermediate 2–3 · high 4–6 · very high 7–10 → 5-yr OS **93** / **79** / **63** / **23%** (chemo era; targeted agents narrow the gaps, esp. for IGHV).

GENETICS (DÖHNER HIERARCHY)

Lesion	Meaning
del(13q) sole *~36% (any ~55%)*	Most favorable; median OS 133 mo (chemo era)
Trisomy 12 *~16%*	Intermediate; often atypical (CD23-dim) phenotype
del(11q) *~18%*	ATM; bulky nodes; adverse with chemo — **largely overcome by BTKi**
del(17p) / TP53 mut *~7% at dx; more at relapse*	Chemo-refractory; **never FCR/BR**; continuous BTKi preferred
IGHV unmutated *~40% at dx*	Shorter remissions after fixed-duration and chemo; no plateau with FCR

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FIRST LINE — EVIDENCE

Regimen · trial	Population · result · the catch
Zanubrutinib · SEQUOIA	≥65 or comorbid, vs BR: 5-yr median PFS **NR vs 44.1 mo, HR 0.29**; **OS no different**. **Arm C del(17p)** (single-arm): 18-mo PFS 88.6%
Acalabrutinib ± O · ELEVATE-TN	Older/comorbid, vs Clb-O: 6-yr PFS **78% (A-O)** / **62% (A)** / **17%**; **OS HR 0.62** A-O vs Clb-O. Only BTKi trial where adding anti-CD20 helps
Ibrutinib · RESONATE-2, A041202, E1912	RESONATE-2 (≥65): **median PFS 8.9 vs 1.3 yr** vs chlorambucil. A041202 (≥65): beat BR, **rituximab adds nothing**; AF 18%, HTN 55%. E1912 (≤70): IR beat FCR, **OS HR 0.47**
Pirtobrutinib · BRUIN CLL-313	No del(17p), vs BR: PFS **HR 0.20**, 24-mo PFS 93.4 vs 70.7%. **FDA 1L approval Oct 2026** (no del17p). CLL-314: ORR noninferior to ibrutinib
VenO ×12 · CLL14	Unfit (CIRS >6 or CrCl <70), vs Clb-O: median PFS **76.2 vs 36.4 mo**, HR 0.40; OS NS. Shorter PFS with **del(17p), IGHV-U, nodes ≥5 cm**
CLL13 / GAIA *fit, no TP53*	GV / GIV / RV vs FCR-BR: 5-yr PFS **69.8** / **81.3** / **57.4 vs 50.7%**. **VenR not better than CIT** — obinutuzumab is the 1L partner
AV ± O · AMPLIFY	Fit, no del17p/TP53, vs FCR/BR: 36-mo PFS **AV 76.5%** / **AVO 83.1%** / **66.5%**; AV HR 0.65. AVO: more COVID deaths, lower OS. **FDA Feb 2026 for AV**
Ibrutinib + venetoclax · GLOW, CAPTIVATE	GLOW (older/unfit, vs Clb-O): **OS HR 0.46**. CAPTIVATE (≤70): 24-mo PFS 95%; TP53 36-mo PFS 81%. **EU-approved, not FDA-labeled**
FLAIR *MRD-guided I+V*	Fit ≤75: I+V for 2× the time to uMRD (max 6 yr) vs FCR: 5-yr PFS **93.9 vs 58.1%**, **OS HR 0.26**; also beat ibrutinib alone (OS HR 0.41). Cardiac SAEs 10.7%
CLL17 *fixed vs continuous*	VenO vs I+V vs continuous ibrutinib: 3-yr PFS **81.1** / **79.4** / **81.0%** — **both fixed-duration arms noninferior**. PB uMRD 73.3 / 47.2 / 0%
FCR *now niche*	MDACC 12.8-yr PFS **53.9% if IGHV-mutated** (plateau) vs 8.7% unmutated. Beaten by targeted regimens; therapy-related myeloid neoplasms. **Never for TP53 or IGHV-U**

Fixed vs continuous: equal 3-yr PFS (CLL17). Fixed = drug holiday, less cumulative cardiac/bleeding toxicity, needs TLS ramp-up. Continuous = simplest, preferred for TP53; indefinite cost, AF/HTN. **Obinutuzumab, not rituximab**, with 1L venetoclax.

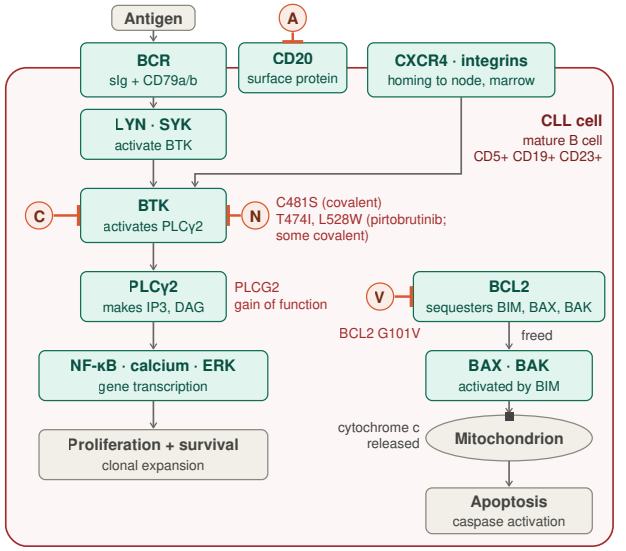
RELAPSED / REFRACTORY — EVIDENCE

Regimen · trial	Result · caveat
VenR ×2 yr · MURANO	vs BR: median PFS **54.7 vs 17.0 mo**; 7-yr OS 69.6 vs 51.0%. VenR retreatment: ORR 72%. Mostly BTKi-naive
Acalabrutinib vs ibrutinib · ELEVATE-RR	del17p/del11q: median PFS **38.4 mo both** (noninferior); less AF, HTN, bleeding
Zanubrutinib vs ibrutinib · ALPINE	**PFS superior, HR 0.65**, incl del17p/TP53; less AF, fewer fatal cardiac events
Pirtobrutinib · BRUIN CLL-321	All post-cBTKi, vs idelalisib-R/BR: median PFS **14.0 vs 8.7 mo**, HR 0.54. **Full approval Dec 2025**: R/R after a cBTKi (BCL2i not required)
Pirtobrutinib-VR · BRUIN CLL-322	Fixed duration vs VR: 24-mo PFS **87 vs 72%**, HR 0.55. Not FDA-approved as of Oct 2026
Liso-cel · TRANSCEND CLL 004	After BTKi + venetoclax: CR/CRi **18%**, ORR 43%; median PFS 11.9 mo. CRS 85%, neurotox 45% (gr 3 18%). **Accelerated approval** after BTKi + BCL2i
Allo-HCT	Only curative option; for fit patients with double-exposed/refractory disease or Richter in remission
Not approved in CLL	BTK degraders, sonrotoclax — trials only (sonrotoclax approved in MCL)

RICHTER TRANSFORMATION

- **~2–10% of CLL (~0.5–1%/yr).** Suspect with **rapid, asymmetric node growth, rising LDH, B symptoms, hypercalcemia**, new extranodal disease
- **PET-CT → biopsy the hottest lesion.** SUVmax ≥10 flags Richter (OS 6.9 vs 56.7 mo); **core or excisional, not FNA** (FNA often nondiagnostic)
- **Clonally related** (same IGHV as the CLL; more TP53 disruption) median OS **14.2 mo** vs **unrelated 62.5 mo** — unrelated behaves like de novo DLBCL
- **Treatment:** R-CHOP-type induction → **allo-HCT in responders**; CD19 CAR-T or bispecifics on trial/off-label. Hodgkin type: ABVD-type. No Richter-specific FDA approval. Median OS overall ~10 mo

PATHOPHYSIOLOGY



C ibrutinib · acalabrutinib · zanubrutinib: covalent BTK inhibitors (bind C481)
N pirtobrutinib: noncovalent BTK inhibitor
V venetoclax: BCL2 inhibitor
A obinutuzumab · rituximab: anti-CD20 antibodies (immune-mediated killing)

C481S blunts all covalent BTK inhibitors; pirtobrutinib binds BTK without C481. Venetoclax targets BCL2, not BCR signaling, so BTK or PLCG2 mutations do not affect it.

RESISTANCE AND SEQUENCING

- **Ibrutinib progression: BTK C481 or PLCG2 mutations in ~85%.** Covalent BTKis share C481 binding → **don't swap cBTKi for progression**; **pirtobrutinib (noncovalent) binds C481-mutant BTK**
- On pirtobrutinib, **non-C481 BTK mutations (T474I, L528W)** emerge; these also cross-resist zanubrutinib/acalabrutinib. Venetoclax resistance: **BCL2 G101V**
- **Intolerance ≠ resistance:** ibrutinib AF/bleeding → switch to acalabrutinib or zanubrutinib
- **Keep the BTKi going until the next therapy starts** — stopping can cause a rapid disease flare
- **Retest FISH + TP53 at every relapse**; new adenopathy with a high LDH → think Richter

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REGIMEN DOSING

Regimen	Dosing
Acalabrutinib	**100 mg BID**, continuous; A-O: obinutuzumab C2–7. **Tablets OK with PPIs** (old capsules: avoid)
Zanubrutinib	**160 mg BID or 320 mg daily**, continuous
Ibrutinib	**420 mg daily**, continuous
Pirtobrutinib	**200 mg daily**, continuous
Venetoclax ramp-up	**20 → 50 → 100 → 200 → 400 mg**, one step per week (5 weeks), with TLS prophylaxis (table)
VenO *CLL14, 12 × 28-d cycles*	Obinutuzumab **100 mg C1D1, 900 mg C1D2, 1000 mg C1D8 + D15**, then 1000 mg D1 C2–6. Venetoclax ramp **starts C1D22**, 400 mg from C3, **stop after C12**
AV *AMPLIFY, 14 cycles*	Acalabrutinib 100 mg BID **C1–14**; venetoclax ramp-up **from C3**, 400 mg through C14. (AVO adds obinutuzumab C2–7 — not in US label)
I+V *GLOW / CAPTIVATE*	Ibrutinib 420 mg **×3 cycles lead-in** (debulks: high-TLS-risk 21% → 1%), then + venetoclax ramp to 400 mg **×12 cycles** (15 total)
VenR *MURANO*	Ramp-up 5 wk, then rituximab **375 mg/m² C1, 500 mg/m² C2–6** (q28d); venetoclax 400 mg for **24 mo from rituximab C1D1**
Liso-cel	Flu/Cy lymphodepletion → **90–110×10^6 CAR+ T cells**; REMS removed; stay near center, no driving ~2 wk

VENETOCLAX TLS RISK (LABEL)

Risk	Criteria → prophylaxis · monitoring
Low	All nodes **<5 cm and ALC <25** → oral fluids 1.5–2 L/d + **allopurinol** (start 2–3 d before) · outpatient; labs **pre, 6–8 h, 24 h** at the first 20 and 50 mg doses
Medium	Any node **5 to <10 cm or ALC ≥25** → oral ± IV fluids + allopurinol · same labs; **consider admission if CrCl <80**
High	Any node **≥10 cm, or ALC ≥25 + any node ≥5 cm** → oral + **IV fluids 150–200 mL/h**, allopurinol, **rasburicase if uric acid high** · **admit for 20 and 50 mg**; labs 0, 4, 8, 12, 24 h

Labs = K, phos, Ca, uric acid, creatinine — **correct before the next dose**. Debulk first (BTKi lead-in in GLOW/AMPLIFY; obinutuzumab in VenO). **No urine alkalinization.**

BTK INHIBITORS COMPARED

Agent	Binding · signature toxicity
Ibrutinib	Covalent (C481), 1st gen · **AF** (16.0% in ELEVATE-RR), **HTN**, ventricular arrhythmia, bleeding, arthralgia
Acalabrutinib	Covalent, more selective · **headache** early; AF 9.4% vs 16.0%, HTN 9.4 vs 23.2% (vs ibrutinib)
Zanubrutinib	Covalent, more selective · **neutropenia**; AF 2.5 vs 10.1% vs ibrutinib (ALPINE interim)
Pirtobrutinib	**Noncovalent** · works with C481 mutations; AF/flutter 3.1% (label, pooled); infections, rash

BTKI TOXICITY MANAGEMENT

- **Atrial fibrillation:** rate control (**avoid diltiazem/verapamil** — CYP3A), DOAC if indicated; often switch to a 2nd-gen BTKi or to venetoclax
- **Ventricular arrhythmia / sudden death:** ibrutinib label warning; FLAIR excess cardiac deaths. Palpitations/syncope → hold + ECG/Holter; don't rechallenge ibrutinib
- **Hypertension:** 55% all-grade on ibrutinib (A041202) — check BP every visit, treat early
- **Bleeding: no warfarin**; avoid fish oil/vitamin E; caution with DOAC + antiplatelet. **Hold 3–7 days before and after procedures**
- **Headache with acalabrutinib** (31–40%, early, self-limited): acetaminophen + caffeine. Also arthralgia, rash, diarrhea, bruising
- **CYP3A:** acalabrutinib — avoid strong inhibitors, 100 mg daily with moderate · zanubrutinib — 80 mg daily (strong), 80 mg BID (moderate) · pirtobrutinib — minus 50 mg with strong · **venetoclax with posaconazole 70 mg**, other strong 100 mg, moderate ≥50% cut; strong inhibitors contraindicated during ramp-up

DISEASE COMPLICATIONS

- **Autoimmune cytopenias ~5–10%: AIHA most common** (warm IgG), ITP ~2%, **PRCA** and autoimmune granulocytopenia <1%. Distinguish from marrow failure (retic, haptoglobin, DAT, marrow). **Steroids ± rituximab**; IVIG for ITP bleeding. **Steroid-refractory autoimmune cytopenia = iwCLL indication** — treat the CLL (BTKi or VenO; **avoid fludarabine monotherapy**)
- **Infection is the leading cause of death:** hypogammaglobulinemia + T-cell and neutrophil dysfunction. **IVIG if IgG <500 mg/dL + recurrent serious bacterial infections** (fewer infections, no survival gain)
- **Vaccines:** before therapy when possible; **no live vaccines**; **recombinant zoster** (2 doses), pneumococcal conjugate, influenza, COVID, RSV — responses blunted on BTKi/anti-CD20
- **Prophylaxis: anti-HBc+ → entecavir** with anti-CD20 (HBV reactivation). **PJP + HSV/VZV** with FCR/BR, idelalisib (label) and after CAR-T; individualize on BTKi/venetoclax
- **Second cancers: skin SCC, BCC, melanoma, Merkel** markedly increased **→ annual skin exam**, sun protection. Acalabrutinib label: second primaries 16% (non-melanoma skin 9%)

CHOOSING FIRST LINE — PATIENT FACTORS

Situation	Lean toward
del(17p) / TP53 mutation	**Continuous zanubrutinib or acalabrutinib**; VenO if a BTKi is contraindicated (expect shorter PFS)
IGHV-mutated, no TP53	**Fixed duration** (VenO or AV) — longest treatment-free intervals; FCR no longer needed
IGHV-unmutated, no TP53	Either: fixed duration (VenO, AV, I+V) or continuous BTKi — CLL17 equal PFS; MRD-guided I+V (FLAIR) for fit
AF, heart failure, anticoagulation, bleeding risk	**VenO**; if a BTKi is needed, zanubrutinib or acalabrutinib, never ibrutinib first
Renal impairment, bulky high-TLS-risk, can't do TLS labs	**Continuous BTKi**, or a BTKi lead-in before venetoclax (AV, I+V)
Frail / comorbid	Acalabrutinib, zanubrutinib or VenO (CLL14 enrolled CIRS >6)

RESPONSE AND MRD

- **iwCLL CR:** ALC <4×10^9/L, nodes <1.5 cm, no hepatosplenomegaly, no B symptoms, ANC ≥1.5, plt ≥100, Hb ≥11 untransfused. Most patients on a BTKi reach PR, not CR — still durable
- **BTKi redistribution lymphocytosis** in the first weeks (cells leave nodes) is expected, **not progression**
- **uMRD = <10^{-4}** (1 CLL cell in 10,000) by flow or NGS. Predicts PFS after fixed-duration therapy (MURANO, GLOW, CLL14); **meaningless on continuous BTKi** (CLL17 0%). MRD-guided duration is a trial/FLAIR concept, not routine

ON-TREATMENT MONITORING

- **Venetoclax neutropenia** (gr 3–4 52.8% in CLL14): **G-CSF**, hold/dose-reduce per label rather than stopping; check CBC each cycle
- **Obinutuzumab infusion reactions** (40% with Clb-O vs 13% when given with acalabrutinib in ELEVATE-TN): split first dose 100/900 mg, premedicate (steroid, acetaminophen, antihistamine), hold antihypertensives the morning of dose 1
- **BTKi:** CBC, BP, pulse every visit; ask about palpitations, bruising, infections. Long-term: **~20% of ibrutinib patients stop for toxicity** (E1912: 21.9% by 5.8 yr)
- **Off therapy** (fixed duration or watch-and-wait): CBC + exam q3–6 mo; **imaging only for symptoms or before re-treatment**